

Cluster 1

To view the latest Map of Science clusters and data, visit the live Map at sciencemap.eto.tech.

Subjects

Research fields: Chemistry, Materials science, Physics

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Nuclear chemistry, Inorganic chemistry, Physical chemistry

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: corrosion inhibitors, mild steel corrosion, mild steel, inhibition efficiency, carbon steel

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

0.15% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to cybersecurity

Vital signs

5364 recent articles in the cluster

5364 articles in this cluster were published in the five year period ending in 2023.

Average article is **10.00** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **61.22th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 61.22% of other clusters.

Citation rating: **65.53th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 65.53% of all other articles published in the same year.

95.84% of articles are in English

Expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 4997 of 5364 articles. 4186 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
India	1016	4.26
China	979	5.41
Saudi Arabia	831	6.44
Morocco	707	5.66
Egypt	602	4.88
Nigeria	509	2.46
South Korea	242	7.29
Iraq	233	3.52
Turkey	219	7.10
Iran	203	6.02

Top funding countries

Data available for 1496 of 5364 articles. 1325 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	584	7.03
Saudi Arabia	294	6.72
India	145	5.83
South Korea	105	8.01
Brazil	68	3.77
Egypt	63	5.55
Iran	59	8.06
Malaysia	40	7.69
South Africa	33	6.16
United States	27	6.54

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[Plant Extracts as Green Corrosion Inhibitors for Different Metal Surfaces and Corrosive Media: A Review](#)

2020: Processes. 254 citations.

[Papaya leaves extract as a novel eco-friendly corrosion inhibitor for Cu in H2SO4 medium](#)

2020: Journal of colloid and interface science. 350 citations.

[Fabrication of environmentally friendly Losartan potassium film for corrosion inhibition of mild steel in HCl medium](#)

2020: Chemical Engineering Journal. 371 citations.

[Pyracantha fortuneana alcohol extracts as biodegradable corrosion inhibitors for copper in H2SO4 media](#)

2024: Journal of Molecular Liquids. 61 citations.

[Principles and theories of green chemistry for corrosion science and engineering: design and application](#)

2024: Green Chemistry. 20 citations.

[Investigation of adsorption, corrosion inhibition, synergistic inhibition effect and stability studies of Rheum ribes leaf extract on mild steel in 1 M HCl solution](#)

2023: Journal of the Taiwan Institute of Chemical Engineers / Elsevier. 66 citations.

[The use of methanol extract of Rheum Ribes \(İşgin\) flower as a natural and promising corrosion inhibitor for mild steel protection in 1 M HCl solution](#)

2023: Journal of Industrial and Engineering Chemistry. 76 citations.

[Dardagan Fruit extract as eco-friendly corrosion inhibitor for mild steel in 1 M HCl: Electrochemical and surface morphological studies](#)

2020: Journal of the Taiwan Institute of Chemical Engineers. 195 citations.

[Insight into anti-corrosion nature of Betel leaves water extracts as the novel and eco-friendly inhibitors](#)

2020: Journal of colloid and interface science. 232 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

[Corrosion Inhibition Applications of Natural and Eco-Friendly Corrosion Inhibitors on Steel in the Acidic Environment: An Overview](#)

2021: Arabian Journal for Science and Engineering. 33 citations.

[Synthesis, structural characterization, thermal analysis and anticorrosion properties of novel AZO-based imidazo\[1,2-a\]pyridine derivatives: Experimental study and theoretical approaches](#)

2024: Inorganic Chemistry Communications. 0 citations.

[Plant extracts as corrosion inhibitors for mild steel in H2SO4 and H3PO4 media – Review II](#)

2022: International Journal of Corrosion and Scale Inhibition. 10 citations.

[Investigation of inhibitive performance of Betahistine dihydrochloride on mild steel in 1 M HCl solution](#)

2021: Journal of Molecular Liquids. 38 citations.

[Mechanistic evaluation of adsorption and corrosion inhibition capabilities of novel indoline compounds for oil well/tubing steel in 15% HCl](#)

2021: Chemical Engineering Journal. 78 citations.

[Corrosion inhibition of mild steel in acidic environments: Mechanistic insights and protective effects of azo-cum inhibitor](#)

2024: Journal of environmental chemical engineering. 6 citations.

[Chemical medicines \(drugs\) as green corrosion inhibitors](#)

2022: The Royal Society of Chemistry eBooks. 1 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[Plant Extracts as Green Corrosion Inhibitors for Different Metal Surfaces and Corrosive Media: A Review](#)

2020: Processes. 254 citations.

[Papaya leaves extract as a novel eco-friendly corrosion inhibitor for Cu in H2SO4 medium](#)

2020: Journal of colloid and interface science. 350 citations.

[Multidimensional insights into the corrosion inhibition of 3,3-dithiodipropionic acid on Q235 steel in H2SO4 medium: A combined experimental and in silico investigation.](#)

2020: Journal of colloid and interface science. 236 citations.

[Fabrication of environmentally friendly Losartan potassium film for corrosion inhibition of mild steel in HCl medium](#)

2020: Chemical Engineering Journal. 371 citations.

[Self-assembling anchored film basing on two tetrazole derivatives for application to protect copper in sulfuric acid environment](#)

2020: 材料科学技术学报. 251 citations.

[Challenges and advantages of using plant extract as inhibitors in modern corrosion inhibition systems: Recent advancements](#)

2020: Journal of Molecular Liquids. 212 citations.

[Insight into anti-corrosion nature of Betel leaves water extracts as the novel and eco-friendly inhibitors](#)

2020: Journal of colloid and interface science. 232 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 5302 out of 5364 articles. 1457 of these articles came from these sources.

Source	Articles (last five years)
Journal of Molecular Liquids	384
Journal of Molecular Structure	188
International Journal of Electrochemical Science	175
International Journal of Corrosion and Scale Inhibition	161
Journal of Bio- and Tribo-Corrosion	138
Colloids and Surfaces A: Physicochemical and Engineering Aspects	130
Research Square (Research Square)	80
Journal of Adhesion Science and Technology	69
ACS omega	68
Scientific reports	64

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 5333 of 5364 articles. 740 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
A. Zarrouk	177	12.24
Lei Guo	137	30.56
F. Benhiba	120	25.17
Chandrabhan Verma	115	19.28
M.A. Quraishi	114	10.75
A. S. Fouda	108	4.37
Ismail Warad	107	8.78
Abdelkader Zarrouk	103	9.57
Hassane Lgaz	100	16.70
Savaş Kaya	93	9.75

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 5032 of 5364 articles. 1242 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
Université Ibn-Tofail - Morocco	270	7.35
Mohammed V University - Morocco	245	7.06
King Fahd University of Petroleum and Minerals - Saudi Arabia	238	7.7
Mansoura University - Egypt	191	4.74
Egyptian Petroleum Research Institute - Egypt	150	5.31
Sidi Mohamed Ben Abdellah University - Morocco	134	7.8
Tongren University - China	133	10.96
Université Ibn Zohr - Morocco	127	6.23
Southwest Petroleum University - China	124	5.48
Umm al-Qura University - Saudi Arabia	112	5.97

Organization types

This table covers the 4743 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 5364 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
0.77%	52.23%	4.41%	1.61%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 143 out of 5364 articles. 75 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
China National Petroleum Corporation (China)	32	7.72
Sinopec (China)	15	2.09
China National Offshore Oil Corporation (China)	7	1.93
Saudi Aramco (Saudi Arabia)	7	6.84
Tata Steel (India)	4	3.58
Beijing Chemical Industry Research Institute (China)	4	4.75
Petronas (Malaysia)	4	3.42
Pertamina (Indonesia)	3	2.17
Bharat Petroleum (India)	3	9.08
Yonker Environmental Protection (China)	3	53.33

Top funders

Data available for 2178 out of 5364 recent articles. 840 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	453	7.34
King Saud University	88	7.71
National Research Foundation of Korea - South Korea	61	8.42
National Council for Scientific and Technological Development - Brazil	58	3.85
Ministry of Education - China	54	5.26
King Saud University - Saudi Arabia	48	5.06
University Grants Commission - India	47	5.99
King Fahd University of Petroleum and Minerals - Saudi Arabia	46	8.52
Ministry of Education - Brazil	46	3.78
City of Chongqing - China	45	10.73

Funder types

This table covers the 1381 articles in this cluster from the last five years with data about their funding organization types (out of 5364 articles total).

Commercial	Education	Nonprofit	Government
0.92%	36.41%	2.93%	59.73%

Patents and industry

363 patents cite articles in this cluster (**95.13th** percentile among all clusters)

2.7% of articles in the cluster are cited by patents

0.71% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
INHIBITING CORROSION IN GAS WELLS	8
CORROSION INHIBITOR COMPOSITION AND METHODS OF INHIBITING CORROSION DURING ACID PICKLING	8
CORROSION INHIBITORS	7
CORROSION INHIBITION	18
AQUEOUS ACIDIC COMPOSITIONS WITH A CORROSION INHIBITOR	17
FORMULATION COMPRISING 2-SUBSTITUTED BENZIMIDAZOLE OR 2-SUBSTITUTED IMIDAZOLE CORROSION INHIBITORS	13
WATER-SOLUBLE PYRAZOLE DERIVATIVES AS CORROSION INHIBITORS	11
SYSTEMS, TOOLS, AND METHODS FOR FLOW ASSURANCE MONITORING	10
PURINE-BASED CORROSION INHIBITORS	10

Top patent assignees

Patent assignee	Same-cluster citations
KING FAHD UNIVERSITY OF PETROLEUM AND MINERALS	40
ECOLAB (UNITED STATES)	34
SCHLUMBERGER TECHNOLOGY CORPORATION	32
SCHLESWIG HOLSTEIN NETZ AG	32
HAITHAM RAKAD MOHAMMAD AL HATAMLEH	17
THE BOEING COMPANY	17
SCHLUMBERGER HOLDINGS	16
エコラボ ユーエスエー インコーポレイティド	16
艺康美国股份有限公司	14

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
10613	318	corrosion inhibition, copper surface, copper corrosion, inhibition efficiency, adsorption
11070	276	epoxy coating, corrosion protection, graphene composite coatings, electrochemical impedance spectroscopy, EIS
20750	215	corrosion protection, composite coating, LDHs corrosion inhibitor, self-healing coatings, LDH film
5092	192	corrosion inhibitors, carbon steel, Localized corrosion, steel surface, pipeline steel
25798	184	corrosion inhibitor, concrete pore solution, simulated concrete pore, reinforced concrete, steel rebar
72619	157	Mild Steel Corrosion, mild steel, corrosion inhibitor, inhibition efficiency, Langmuir adsorption isotherm
3999	93	microbiologically influenced corrosion, Microbial corrosion, SRB MIC, Carbon steel MIC, Stainless steel corrosion
73168	91	corrosion inhibitor, ionic liquids, carbon steel, electrochemical impedance spectroscopy, PANI
47582	77	corrosion inhibition, mild steel, mild steel corrosion, carbon steel, steel surface
6779	76	corrosion protection, composite coating, PPy, carbon steel, prepared PANI coating

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
10613	541	corrosion inhibition, copper surface, copper corrosion, inhibition efficiency, adsorption
5092	257	corrosion inhibitors, carbon steel, Localized corrosion, steel surface, pipeline steel
11070	225	epoxy coating, corrosion protection, graphene composite coatings, electrochemical impedance spectroscopy, EIS
5552	200	density functional theory, Conceptual density functional, chemical reactivity, reactivity descriptors, conceptual DFT
20750	74	corrosion protection, composite coating, LDHs corrosion inhibitor, self-healing coatings, LDH film
84	70	carbon quantum dots, Fluorescent carbon dots, CDs, carbon dot synthesis, Polymer Dots
25798	70	corrosion inhibitor, concrete pore solution, simulated concrete pore, reinforced concrete, steel rebar
5668	70	Gemini surfactants, Cationic gemini, Gemini cationic, surface tension, critical micelle concentration
10608	57	conversion coating, corrosion resistance, Electrochemical Impedance Spectroscopy, X-ray photoelectron spectroscopy, cerium conversion
6779	54	corrosion protection, composite coating, PPy, carbon steel, prepared PANI coating

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
mwang87/MergePolarity	1	1